Amazon RDS (Relational Database Services)

Amazon Relational Database Service (Amazon RDS) makes database configuration, management, and scaling easy in the cloud. Automate tedious tasks such as hardware provisioning, database arrangement, patching, and backups – cost-effectively and proportionate to your needs.

RDS is available on various database instances which are optimized for performance and memory, providing six familiar database engines including Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle. database, and SQL server. By leveraging the AWS Database Migration Service, you can easily migrate or reproduce your existing databases to Amazon RDS. Visit Amazon's RDS page.

https://allcode.com/top-aws-services/?fbclid=IwAR3mF-EkBTIwx3G2nVSEQqRw5x38Da8v9aEfVnsASxwGvExLJJv6BdQH_U8

Amazon RDS Partner

AllCode is an AWS RDS Service Delivery Partner (SDP). AllCode will help you implement AWS RDS to eliminate hardware provisioning and lowering operational overhead, allowing your team to focus on optimizing applications for efficiency, data integrity, and security.
AllCode has extensive experience in RDS and in traditional RDBMS. We possess the original software developers of Embarcadero's DBArtisan and RapidSQL. We even have the original architect of the Embarcadero products, none other than Mr. Wire Science himself, Chris Hanson.
With over 20 years of experience, our CEO wrote the UI and backend debuggers for the Oracle PL/SQL, Sybase and SQL Server T-SQL in MFC interfacing with the legacy OCI and COM

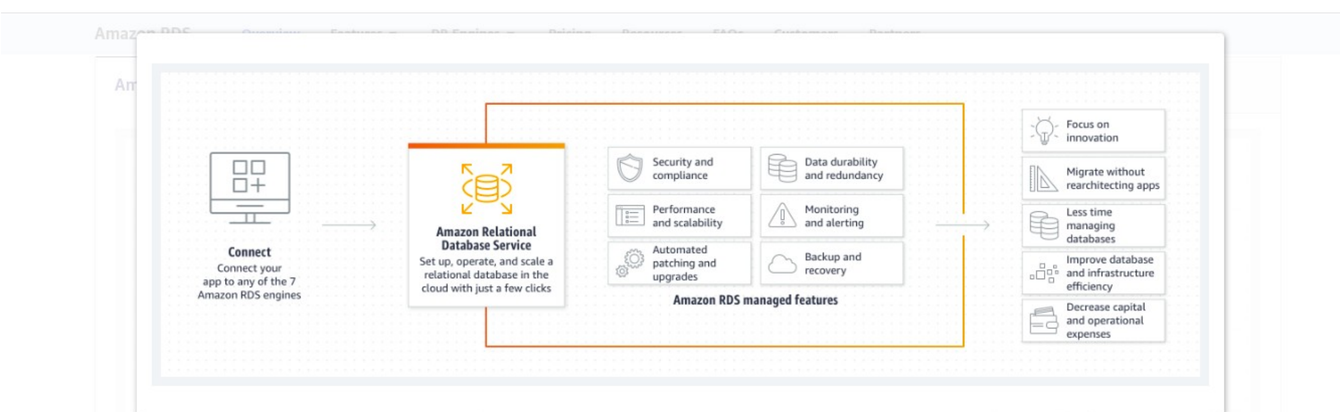
that reside in DBArtisan and RapidSQL. We're such database gear heads that we're not quite sure why you'd ever trust your data model design to anyone else.
We specialize in creating your data model via an ER Modeler, deploying your DDL to the appropriate RDS instance, tuning your SQL to ensure query response time, and operating your RDS instance to ensure 24/7 availability.

https://allcode.com/amazon-web-services/aws-rds-consulting-partner/

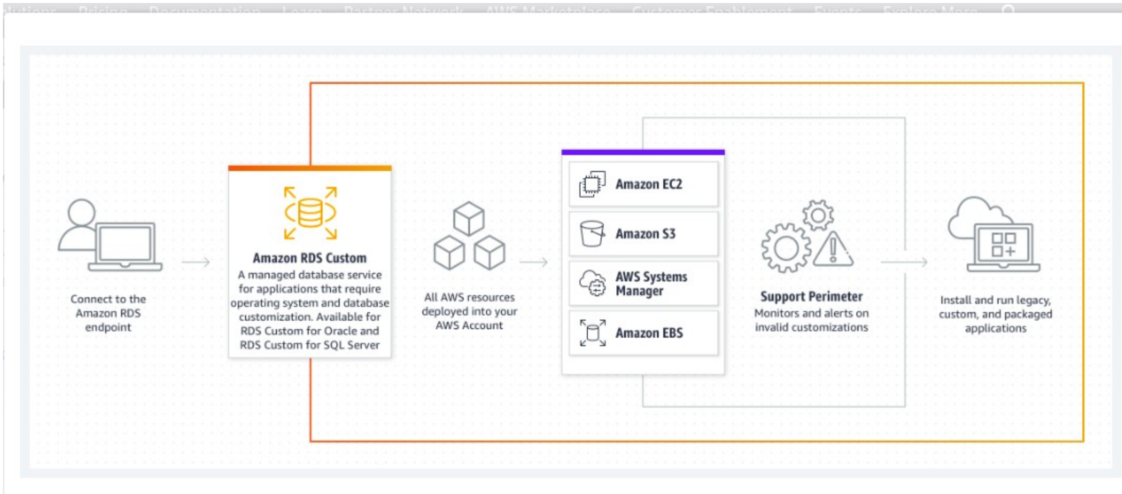
How it works

Amazon Relational Database Service (Amazon RDS) is a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud. Choose from seven popular engines — Amazon Aurora with MySQL compatibility, Amazon Aurora with PostgreSQL compatibility, MySQL, MariaDB, PostgreSQL, Oracle, and SQL Server — and deploy on-premises with Amazon RDS on AWS Outposts.

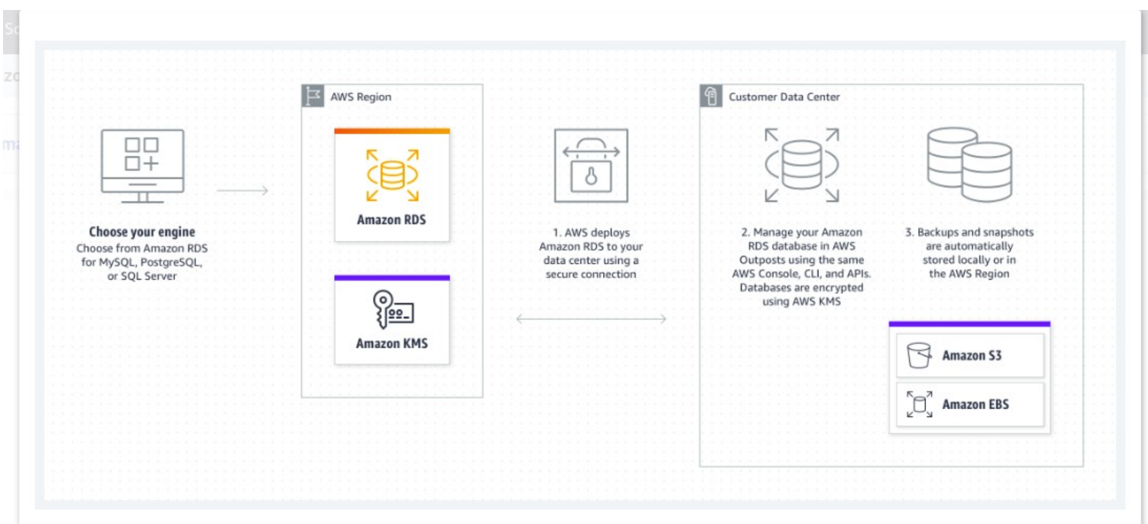
AMAZON RDS



AMAZON RDS CUSTOM



AMAZON RDS ON AWS OUTPOSTS



https://aws.amazon.com/rds/

Amazon RDS Features

Amazon RDS is a managed relational database service that provides you six familiar database engines to choose from, including Amazon Aurora, MySQL, MariaDB, PostgreSQL, Oracle, and Microsoft SQL Server. This means that the code, applications, and tools you already use today with your existing databases can be used with Amazon RDS. Amazon RDS handles routine database tasks, such as provisioning, patching, backup, recovery, failure detection, and repair.

Amazon RDS makes it easy to use replication to enhance availability and reliability for production workloads. Using the Multi-AZ deployment option, you can run mission-critical workloads with high availability and built-in automated failover from your primary database to a synchronously replicated secondary database. Using Read Replicas, you can scale out beyond the capacity of a single database deployment for read-heavy database workloads.

As with all Amazon Web Services, there are no up-front investments required and you pay only for the resources you use.

Lower administrative burden

Easy to use

You can use the AWS Management Console, the Amazon RDS Command Line Interface, or simple API calls to access the capabilities of a production-ready relational database in minutes.

Amazon RDS database instances are pre-configured with parameters and settings appropriate for the engine and class you have selected. You can launch a database instance and connect your application within minutes. DB Parameter Groups provide granular control and fine-tuning of your database.

Automatic software patching

Amazon RDS will make sure that the relational database software powering your deployment stays up-to-date with the latest patches. You can exert optional control over when and if your database instance is patched.

Best practice recommendations

Amazon RDS provides best practice guidance by analyzing configuration and usage metrics from your database instances. Recommendations cover areas such as database engine versions, storage, instance types, and networking. You can browse the available recommendations and perform a recommended action immediately, schedule it for their next maintenance window, or dismiss it entirely.

Performance

General Purpose (SSD) Storage

Amazon RDS General Purpose Storage is an SSD-backed storage option that delivers a consistent baseline of 3 IOPS per provisioned GB and provides the ability to burst up to 3,000 IOPS above the baseline. This storage type is suitable for a broad range of database workloads.

Provisioned IOPS (SSD) Storage

Amazon RDS Provisioned IOPS Storage is an SSD-backed storage option designed to deliver fast, predictable, and consistent I/O performance. You specify an IOPS rate when creating a database instance, and Amazon RDS provisions that IOPS rate for the lifetime of the database instance. This storage type is optimized for I/O-intensive transactional (OLTP) database workloads. You can provision up to 40,000 IOPS per database instance, although your actual realized IOPS may vary based on your database workload, instance type, and database engine choice.

Scalability

Push-button compute scaling

You can scale the compute and memory resources powering your deployment up or down, up to a maximum of 32 vCPUs and 244 GiB of RAM. Compute scaling operations typically complete in a few minutes.

Easy storage scaling

As your storage requirements grow, you can also provision additional storage. The Amazon Aurora engine will automatically grow the size of your database volume as your database storage needs grow, up to a maximum of 64 TB or a maximum you define. The MySQL, MariaDB, Oracle, and PostgreSQL engines allow you to scale up to 64 TB of storage and SQL Server supports up to 16 TB. Storage scaling is on-the-fly with zero downtime.

Read Replicas

Read Replicas make it easy to elastically scale out beyond the capacity constraints of a single DB instance for read-heavy database workloads. You can create one or more replicas of a given source DB instance and serve high-volume application read traffic from multiple copies of your data, thereby increasing aggregate read throughput. Read replicas are available in Amazon RDS for MySQL, MariaDB, PostgreSQL, and Oracle as well as Amazon Aurora.

Availability and durability

Automated backups

The automated backup feature of Amazon RDS enables point-in-time recovery for your database instance. Amazon RDS will back up your database and transaction logs and store both for a user-specified retention period. This allows you to restore your database instance to any second during your retention period, up to the last five minutes. Your automatic backup retention period can be configured to up to thirty-five days.

Database snapshots

Database snapshots are user-initiated backups of your instances stored in Amazon S3 that are kept until you explicitly delete them. You can create a new instance from a database snapshot whenever you desire. Although database snapshots serve operationally as full backups, you are billed only for incremental storage use.

Multi-AZ deployments

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for database instances, making them a natural fit for production database workloads. When you provision a Multi-AZ database instance, Amazon RDS synchronously replicates your data to a standby instance in a different Availability Zone (AZ).

Automatic host replacement

Amazon RDS will automatically replace the compute instance powering your deployment in the event of a hardware failure.

Security

Encryption at rest and in transit

Amazon RDS allows you to encrypt your databases using keys you manage through AWS Key Management Service (KMS). On a database instance running with Amazon RDS encryption, data stored at rest in the underlying storage is encrypted, as are its automated backups, read replicas, and snapshots.

Amazon RDS supports Transparent Data Encryption in SQL Server and Oracle. Transparent Data Encryption in Oracle is integrated with AWS CloudHSM, which allows you to securely generate, store, and manage your cryptographic keys in single-tenant Hardware Security Module (HSM) appliances within the AWS cloud.

Amazon RDS supports the use of SSL to secure data in transit.

Network isolation

AWS recommends that you run your database instances in Amazon VPC, which allows you to isolate your database in your own virtual network and connect to your on-premises IT infrastructure using industry-standard encrypted IPsec VPNs. You can configure firewall settings and control network access to your database instances.

Resource-level permissions

Amazon RDS is integrated with AWS Identity and Access Management (IAM) and provides you the ability to control the actions that your AWS IAM users and groups can take on specific Amazon RDS resources, from database instances through snapshots, parameter groups, and option groups. You can also tag your Amazon RDS resources and control the actions that your IAM users and groups can take on groups of resources that have the same tag and associated value. For example, you can configure your IAM rules to ensure developers are able to modify "Development" database instances, but only Database Administrators can make changes to "Production" database instances.

Manageability

Monitoring and metrics

Amazon RDS provides Amazon CloudWatch metrics for your database instances at no additional charge. You can use the RDS Management Console to view key operational metrics, including compute/memory/storage capacity utilization, I/O activity, and instance connections. Amazon RDS also provides Enhanced Monitoring, which provides access to over 50 CPU, memory, file system, and disk I/O metrics, and Performance Insights, an easy-to-use tool that helps you quickly detect performance problems.

Event notifications

Amazon RDS can notify you via email or SMS text message of database events through Amazon SNS. You can use the AWS Management Console or the Amazon RDS APIs to subscribe to over 40 different database events associated with your database instances.

Configuration governance

Amazon RDS integrates with AWS Config to support compliance and enhance security by recording and auditing changes to the configuration of your DB instance, including parameter groups, subnet groups, snapshots, security groups, and event subscriptions.

Cost-effectiveness

Pay only for what you use

There is no up-front commitment with Amazon RDS; you simply pay a monthly charge for each database instance that you launch. And, when you’re finished with a database instance, you can easily delete it. To see more details, visit the Amazon RDS Instance Types page and the Amazon RDS Pricing page.

Reserved instances

Amazon RDS Reserved Instances give you the option to reserve a DB instance for a one or three year term and in turn receive a significant discount compared to the On-Demand Instance pricing for the DB instance.

Stop and start

Amazon RDS allows you to easily stop and start your database instances for up to 7 days at a time. This makes it easy and affordable to use databases for development and test purposes, where the database is not required to be running all of the time.

https://aws.amazon.com/rds/features/?pg=ln&sec=be

Amazon RDS Partners

Amazon RDS Partners help you with database monitoring, security, and performance using Amazon RDS database engines, including Amazon Aurora MySQL, Amazon Aurora PostgreSQL, PostgreSQL, MySQL, MariaDB, Oracle, and SQL Server. Amazon RDS Delivery Partners help you set up, operate, and scale a relational database in the cloud. Amazon RDS Ready Partner products have been validated to either provide tooling for Amazon RDS including migration, performance, governance, and monitoring, or support the use of the Amazon RDS platform for applications deployed within a customer's account or provided as SaaS deployed in AWS Partners' Account.

The AWS Service Delivery and AWS Service Ready Programs enable AWS customers to identify AWS Partners experience and a deep understanding of specific AWS services, as well as software solutions. These partners have passed a rigorous technical validation to ensure they are following best practices with Amazon RDS, as well as demonstrated proven customer success.

https://aws.amazon.com/rds/partners/?pg=ln&sec=hs&blog-posts-cards.sort-by=item.additionalFields.modifiedDate&blog-posts-cards.sort-order=desc&partner-solutions-cards.sort-by=item.additionalFields.partnerNameLower&partner-solutions-cards.sort-order=asc&awsf.partner-solutions-filter-partner-type=partner-type%23consulting%7Cpartner-type%23technology&awsf.partner-solutions-filter-product=*all&awsf.partner-solutions-filter-location=*all

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Amazon RDS Pricing

Amazon Relational Database Service (Amazon RDS) is a managed, highly available, and secure database service that makes it simple to set up, operate, and scale databases in the cloud. Amazon RDS is free to try and you pay only for what you use with no minimum fees. You can pay for Amazon RDS using On-Demand or Reserved Instances. Estimate your monthly bill using the AWS Pricing Calculator.

Amazon RDS provides a selection of instance types optimized to fit different relational database use cases. Select one of the Amazon RDS database engines below to view pricing. See Previous Generation Instances for previous instance pricing not listed here.

For Amazon RDS feature-level pricing, see RDS Performance Insights and RDS Proxy pricing pages.

As part of the AWS Free Tier, Amazon RDS helps new AWS customers get started for free with a managed database service in the cloud. Each calendar month, the Amazon RDS Free Tier allows you to use:

- 750 hours of Amazon RDS Single-AZ db.t2.micro, db.t3.micro, and db.t4g.micro Instances usage running MySQL, MariaDB, PostgreSQL databases each month. If running more than one instance, usage is aggregated across instance classes.
- 750 hours of Amazon RDS Single-AZ db.t2.micro Instance usage running Oracle BYOL or SQL Server (running SQL Server Express Edition). Oracle BYOL db.t3.micro Single-AZ Instance usage is also included as part of the Amazon RDS free tier. If running both a db.t2.micro Single-AZ Instance and a db.t3.micro Single-AZ Instance on Oracle BYOL, usage is aggregated across Instance classes.
- 20 GB of General Purpose (SSD) DB storage.
- 20 GB of storage for your automated database backups and any user-initiated DB Snapshots.

https://aws.amazon.com/rds/pricing/?pg=ln&sec=hs

What is a Relational Database?

A relational database is a collection of data items with pre-defined relationships between them. These items are organized as a set of tables with columns and rows. Tables are used to hold information about the objects to be represented in the database. Each column in a table holds a certain kind of data and a field stores the actual value of an attribute. The rows in the table represent a collection of related values of one object or entity. Each row in a table could be marked with a unique identifier called a primary key, and rows among multiple tables can be made related using foreign keys. This data can be accessed in many different ways without reorganizing the database tables themselves.

https://aws.amazon.com/relational-database/

Amazon RDS (Relational Database Service)

What is Amazon RDS?

Amazon Relational Database Service (RDS) is a managed SQL database service provided by Amazon Web Services (AWS). Amazon RDS supports an array of database engines to store and organize data. It also helps with relational database management tasks, such as data migration, backup, recovery and patching.

Amazon RDS facilitates the deployment and maintenance of relational databases in the cloud. A cloud administrator uses Amazon RDS to set up, operate, manage and scale a relational instance of a cloud database. Amazon RDS is not itself a database; it is a service used to manage relational databases.

How does Amazon RDS work?

Databases are used to store large quantities of data that applications can draw on to help them perform various functions. A relational database uses tables to store data. It is called relational because it organizes data points with defined relationships.

Administrators control Amazon RDS with the AWS Management Console, Amazon RDS API calls or the AWS Command Line Interface. They use these interfaces to deploy database instances to which users can apply specific settings.

Amazon provides several instance types with different combinations of resources, such as CPU, memory, storage options and networking capacity. Each type comes in a variety of sizes to suit the needs of different workloads.

RDS users can use AWS Identity and Access Management to define and set permissions for who can access an RDS database.

Amazon RDS features

Amazon RDS features include the following:

Replication. RDS uses the Replication feature to create read replicas. These are read-only copies of database instances that applications use without altering the original production database. Administrators can also enable automatic failover across multiple availability zones through RDS Multi-AZ deployment and with synchronous data replication.

Storage. RDS provides three types of storage:

- **General-purpose solid-state drive (SSD).** Amazon recommends this storage as the default choice.
- **Provisioned input-output operations per second (IOPS).** SSD storage for I/O-intensive workloads.
- **Magnetic.** A lower cost option.

Monitoring. The Amazon CloudWatch service enables managed monitoring. It lets users view capacity and I/O metrics.

Patching. RDS provides patches for whichever database engine the user chooses.

Backups. Another feature is failure detection and recovery. RDS provides managed instance backups with transaction logs to enable point-in-time recovery. Users pick a retention period and restore databases to any time during that period. They also can manually take snapshots of instances that remain until they are manually deleted.

Incremental billing. Users pay a monthly fee for the instances they launch.

Encryption. RDS uses public key encryption to secure automated backups, read replicas, data snapshots and other data stored at rest.

What are the benefits and drawbacks of Amazon RDS?

There are several pros and cons to using Amazon RDS.

Benefits

The main benefit of Amazon RDS is that it helps organizations deal with the complexity of managing large relational databases. Other benefits include the following:

- **Ease of use.** Admins don't need to learn specific database management tools. They also can manage multiple database instances using the management console. RDS is compatible with database engines that users may already be familiar with, such as MySQL and Oracle And it automates manual backup and recovery processes.

- **Cost-effectiveness.** According to AWS, customers only pay for what they use. Also, the time spent maintaining instances is reduced, because maintenance tasks, such as backups and patching, are automated.

- The use of read replicas routes read-heavy traffic away from the main database instance, reducing the workload on that one instance.

- RDS splits up compute and storage so admins can scale them independently.

Drawbacks

Some downsides of using Amazon RDS include the following:

- **Lack of root access.** Because it is a managed service, users do not have root access to the server running RDS. RDS restricts access for certain procedures to those with advanced privileges.

- **Downtime.** Systems must go offline for some patching and scaling procedures. The timing on these processes varies. With scaling, compute resources need a few minutes downtime on average.

Amazon RDS database instances

A database administrator can create, configure, manage and delete an Amazon RDS instance, along with the resources it uses. An Amazon RDS instance is a cloud database environment. Admins can also spin up many databases or schemas; how many depends on the database used.

Amazon RDS limits each customer to a total of 40 database instances per account. AWS imposes further limitations for Oracle and SQL Server instances. With those database instances, a user generally can only have up to 10.

Amazon RDS database engines

An AWS customer can spin up six types of database engines within Amazon RDS:

1. **Amazon Aurora** is a proprietary AWS relational database engine. Amazon Aurora is compatible with MySQL and PostgreSQL.

2. **RDS for MariaDB** is compatible with MariaDB, an open source relational database management system (RDBMS) that's an offshoot of MySQL.

3. **RDS for MySQL** is compatible with the MySQL open source RDBMS.

4. **RDS for Oracle Database** is compatible with several editions of Oracle Database, including bring-your-own-license and license-included versions.

5. **RDS for PostgreSQL** is compatible with PostgreSQL open source object-RDBMS.

6. **RDS for SQL Server** is compatible with Microsoft SQL Server, an RDBMS.

Amazon RDS adds support for major and minor versions of database engines over time. It is designed to allow admins to specify an engine version when they create a database instance. In most cases, Amazon RDS can support developer code, applications and tools that are already in use with existing databases.

AWS provides other database services, including the following:

- Amazon DynamoDB key-value and document database for NoSQL databases;

- Amazon Neptune for graph databases; and

- AWS Database Migration Service to ease database transfers and transformations.

Amazon RDS use cases

Amazon RDS' scalability, security and availability make it useful for a variety of applications. Some possible uses include the following:

Online retailing. These applications manage complex databases that track inventories, transactions and pricing.

Mobile and online gaming. RDS supports developers that need to continuously update these applications and users who need high availability.

Travel applications. Applications like Airbnb take advantage of RDS' ability to simplify time-consuming database administration tasks and automate database replication. Mobile apps like Airbnb also take advantage of RDS' scalable storage capability.

Streaming applications. Applications like Netflix take advantage of RDS' storage scalability as well, and availability of Amazon RDS, which allows them to handle high demand daily.

Finance applications. These applications, like other mobile applications, can use RDS to simplify administrative database tasks and save time and money.

Business-to-business reporting company Enlyft said 6,096 companies were using Amazon RDS in 2021, including The American Red Cross, Penguin Random House and Zendesk. Amazon also reported in 2021 that Airbnb, Intuit and the U.S. Department of Veterans Affairs are among the organizations that use RDS to support their data workloads.

The takeaway

Amazon RDS helps organizations handle relational database management tasks such as migration, backup, recovery and patching. Some of the main features of Amazon RDS are replication, high performance storage and failure detection.

One of the biggest advantages of Amazon RDS is its ease of use. It lets administrators manage multiple database instances without having to learn other database management tools.

These features enable RDS to help organizations cut costs that come from time-consuming database administration tasks and manage the hidden costs that come with using high-performance storage in AWS. Learn more about ways to manage AWS costs.

https://www.techtarget.com/searchaws/definition/Amazon-Relational-Database-Service-RDS

Advantages of Amazon RDS vs. a personal relational database

One of the advantages of developers using Amazon RDS instead of managing their own databases is that it reduces or eliminates their administrative responsibilities.

https://www.techtarget.com/searchaws/answer/Advantages-of-Amazon-RDS-vs-a-personal-relational-database?_gl=1*1blobyj*_ga*MTM2ODkzOTg5My4xNjU5OTUzOTkx*_ga_TQKE4GS5P9*MTY1OTk4MDk3Mi40LjEuMTY1OTk4MTE2NC4w&_ga=2.141854002.1862003466.1659953992-1368939893.1659953991

What is AWS RDS?

AWS RDS is short for Amazon Relational Database Service
It provides affordable relational databases in the cloud, that is easy to use.

- Amazon RDS minimizes relational database management by automation
- Amazon RDS is a Relational Database Cloud Service
- Amazon RDS creates multiple instances for high availability and failovers
- Amazon RDS supports PostgreSQL, MySQL, Maria DB, Oracle, SQL Server, and Amazon Aurora

Relational Databases

- Amazon RDS stores data as tables, records, and fields
- Values from one table can have a relationship to values in other tables. Relationships are a key part of relational databases.
- Relational databases are often used for storing transactional and analytical data
- Relational databases provide stability and reliability for transactional databases

RDS Pricing - Pay as You Go

Amazon RDS is pay as you go. It is comprised of 3 parts:

1. Hosting. You can choose from different types of hosting depending on your need
2. Storage and Operations. Storage is billed per gigabyte per month, and I/O is billed per million-request

3.Data transferred

https://www.w3schools.com/whatis/whatis_aws_rds.asp

Features

New database instances can be launched from the AWS Management Console or using the Amazon RDS APIs.[18] Amazon RDS offers different features to support different use cases. Some of the major features are:

Multi-Availability Zone (AZ) deployment

In May 2010 Amazon announced Multi-Availability Zone deployment support.[19] Amazon RDS Multi-Availability Zone (AZ) allows users to automatically provision and maintain a synchronous physical or logical "standby" replica, depending on database engine, in a different Availability Zone[20] (independent infrastructure in a physically separate location). Multi-AZ database instance can be developed at creation time or modified to run as a Multi-AZ deployment later. Multi-AZ deployments aim to provide enhanced availability and data durability for MySQL, MariaDB, Oracle, PostgreSQL and SQL Server[21] instances and are targeted for production environments.[22] In the event of planned database maintenance or unplanned service disruption, Amazon RDS automatically fails over to the up-to-date standby, allowing database operations to resume without administrative intervention.

Multi-AZ RDS instances are optional and have a cost associated with them. When creating a RDS instance, the user is asked if they would like to use a Multi-AZ RDS instance. In Multi-AZ RDS deployments backups are done in the standby instance so I/O activity is not suspended any time but you may experience elevated latencies for a few minutes during backups.[23]

Read replicas

Read replicas allow different use cases such as to scale in for read-heavy database workloads. There are up to five replicas available for MySQL, MariaDB, and PostgreSQL. Instances use the native, asynchronous replication functionality of their respective database engines.[24] They have no backups configured by default and are accessible and can be used for read scaling.[25] MySQL and MariaDB read replicas can be made writeable again since October 2012;[26] PostgreSQL read replicas do not support it.[25] Replicas are done at database instance level and do not support replication at database or table level.[27]

Performance metrics and monitoring

Performance metrics for Amazon RDS are available from the AWS Management Console or the Amazon CloudWatch API. In December 2015, Amazon announced an optional enhanced monitoring feature that provides an expanded set of metrics for the MySQL, MariaDB, and Aurora database engines.[28]

RDS costs

Amazon RDS instances are priced very similarly to Amazon Elastic Compute Cloud (EC2). RDS is charged per hour and comes in two packages: On-Demand DB Instances[29] and Reserved DB Instances.[29] On-Demand Instances are at an ongoing hourly usage rate. Reserved DB Instances require an up-front, one-time fee and in turn provide a discount on the hourly usage charge for that instance.

Apart from the hourly cost of running the RDS instance, users are charged for the amount of storage provisioned, data transfers and input and output operations performed. AWS have introduced Provisioned Input and Output Operations, in which the user can define how many IO per second are required by their application. IOPS can contribute significantly to the total cost of running the RDS instance.[30]

As part of the AWS Free Tier, the Amazon RDS Free Tier helps new AWS customers get started with a managed database service in the cloud for free. You can use the Amazon RDS Free Tier to develop new applications, test existing applications, or simply gain hands-on experience with Amazon RDS.[31]

Automatic backups

Amazon RDS creates and saves automated backups of RDS DB instances.[23] The first snapshot of a DB instance contains the data for the full DB instance and subsequent snapshots are incremental, maximum retention period is 35 days. In Multi-AZ RDS deployments backups are done in the standby instance so I/O activity is not suspended for any amount of time but you may experience elevated latencies for a few minutes during backups.[23]

Operation

Database instances can be managed from the AWS Management Console, using the Amazon RDS APIs and using aws cli.[18] Since 1 June 2017,[32] you can stop AWS RDS instances from AWS Management Console or AWS CLI for 7 days at a time. After 7 days, it will be automatically started,[32][33] and since September 2018 RDS instances can be protected from accidental deletion.[34] Increase DB space is supported, but not decrease allocated space.[35] Additionally there is at least a six-hour period where new allocation cannot be done.

https://en.wikipedia.org/wiki/Amazon_Relational_Database_Service

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Amazon RDS provides the following specific advantages over database deployments that aren't fully managed:

- You can use the database products you are already familiar with: MariaDB, Microsoft SQL Server, MySQL, Oracle, and PostgreSQL.
- Amazon RDS manages backups, software patching, automatic failure detection, and recovery.
- You can turn on automated backups, or manually create your own backup snapshots. You can use these backups to restore a database. The Amazon RDS restore process works reliably and efficiently.
- You can get high availability with a primary instance and a synchronous secondary instance that you can fail over to when problems occur. You can also use read replicas to increase read scaling.
- In addition to the security in your database package, you can help control who can access your RDS databases by using Amazon Identity and Access Management (IAM) to define users and permissions. You can also help protect your databases by putting them in a virtual private cloud (VPC).

https://docs.amazonaws.cn/en_us/AmazonRDS/latest/UserGuide/Welcome.html
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Amazon RDS (Relational Database Service) is a fully-managed SQL database cloud service that allows to create and operate relational databases. Using RDS you can access your files and database anywhere in a cost-effective and highly scalable way.

Features of Amazon RDS

Amazon RDS has the following features −

- **Scalable** − Amazon RDS allows to scale the relational database by using AWS Management Console or RDS-specific API. We can increase or decrease your RDS requirements within minutes.
- **Host replacement** − Sometimes these situations occur when the hardware of Amazon RDS fails. There is no need to worry, it will be automatically replaced by Amazon.
- **Inexpensive** − Using Amazon RDS, we pay only for the resources we consume. There is no up-front and long-term commitment.
- **Secure** − Amazon RDS provides complete control over the network to access their database and their associated services.
- **Automatic backups** − Amazon RDS backs up everything in the database including transaction logs up to last five minutes and also manages automatic backup timings.
- **Software patching** − Automatically gets all the latest patches for the database software. We can also specify when the software should be patched using DB Engine Version Management.

https://www.tutorialspoint.com/amazon_web_services/amazon_web_services_relational_database_service.htm

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Amazon Relational Database Service (Amazon RDS)

Benefits

- Integrated role-based access control across all AWS services (IAM)

- Comprehensive, cross service API audit logging and security (CloudTrail)
- Integration with other AWS services (24x7 support and consolidated billing)
- Training and architectural patterns/guidance (well architected)
- Tightly integrated with other Amazon Web Services
- Scalable database in the cloud
- Multiple features that enhance reliability for critical production databases
- Easily migrate/replicate your existing databases to Amazon RDS
- Encryption at rest, in transit for many RDS engines
- Accessible through AWS Management Console, AWS RDS CLI, APIs

https://www.digitalmarketplace.service.gov.uk/g-cloud/services/546138398791169

Amazon Relational Database Service(Amazon RDS)

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate and scale relational databases in the cloud. It provides cost-efficient and resizable capabilities while automating time-consuming administration tasks such as hardware provisioning, database setup, patching, and backup. It frees you up to focus on your applications so that you can provide them with the fast performance, high availability, security and compatibility they need.

Amazon RDS is available on multiple database instance types - optimized for memory, performance or I/O - and gives you six familiar database engines to choose from, including Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle Database and SQL Server Huh. You can use the AWS Database Migration Service to migrate easily or replicate your existing databases to Amazon RDS.

What is Amazon RDS?

Amazon Relational Database Service (RDS) is a managed SQL database service provided by Amazon Web Services (AWS). Amazon RDS supports an array of database engines to store and organize data. It also helps in relational database management tasks like data migration, backup, recovery and patching.

Amazon RDS facilitates the deployment and maintenance of relational databases in the cloud. Cloud administrators use Amazon RDS to set up, operate, manage, and scale relational instances of cloud databases. Amazon RDS itself is not a database; It is a service used to manage relational databases.

How does Amazon RDS work?

Databases store large amounts of data that applications can draw upon to help them perform various tasks. A relational database uses tables to store data and is called relational because it organizes data points with defined relationships.

Administrators control Amazon RDS with the AWS Management Console, Amazon RDS API calls, or the AWS command-line interface. They use these interfaces to deploy database instances to which users can apply specific settings.

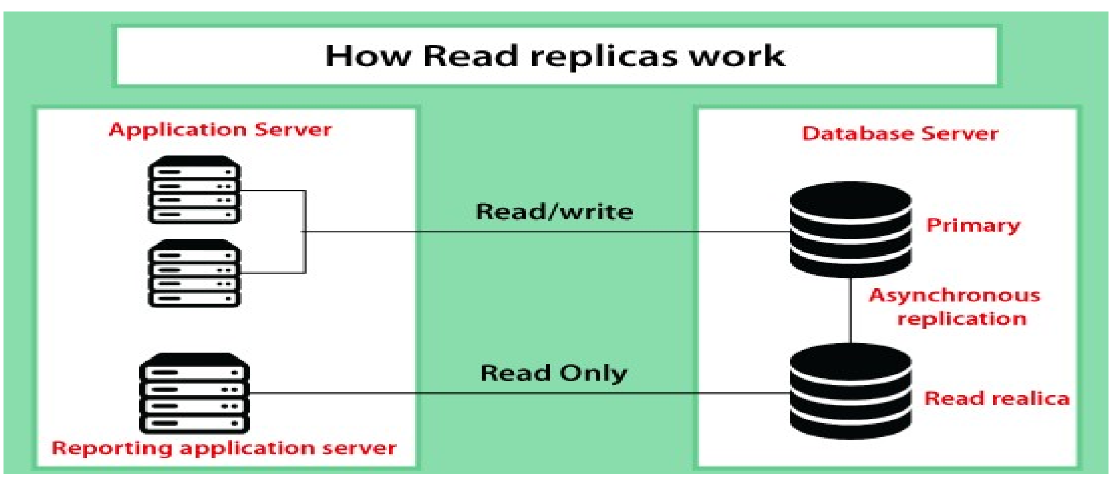
Amazon provides several instance types with different resources, such as CPU, memory, storage options, and networking capability. Each type comes in a variety of sizes to suit the needs of different workloads.

RDS users can use AWS Identity and Access Management to define and set permissions to access RDS databases.

Amazon RDS Features

Amazon RDS features include the following:

Replication. RDS uses the replication feature to create read replicas, and these are read-only copies of the database instances that the application uses without changing the original production database. Administrators can also enable automatic failover across multiple availability zones through RDS multi-edge deployment and synchronous data replication.



RDS provides three types of storage:

A general-purpose solid-state drive (SSD). Amazon recommends this storage as the default choice.

Provisioned input-output operations per second (IOPS). SSD storage for I/O-intensive workloads.

Magnetic. A lower-cost option.

Monitoring. The Amazon **CloudWatch** service enables managed monitoring, and it lets users view capacity and I/O metrics.

Patching. RDS provides patches for whichever database engine the user chooses.

Backups. Another feature is failure detection and recovery. RDS provides managed instance backups with transaction logs to enable point-in-time recovery. Users pick a retention period and restore databases to any time during that period. They also can manually take snapshots of instances that remain until they are manually deleted.

RDS lets users specify the time and duration of the backup processes. They also can choose how long to retain backups and snapshots.

- ❍**Incremental billing.** Users pay a monthly fee for the instances they launch.
- ❍RDS uses public-key encryption to secure automated backups, read replicas, data snapshots and other data stored at rest.

What are the benefits and drawbacks of Amazon RDS?

There are several pros and cons to using Amazon RDS.

- ❍**Easy to administer:** Amazon RDS makes it easy to go from project conception to deployment. Use the Amazon RDS Management Console, the AWS RDS Command-Line Interface, or simple API calls to access the capabilities of a production-ready relational database in minutes. No need for infrastructure provisioning and no need for installing and maintaining database software.
- ❍**Highly scalable:** You can scale your database's compute and storage resources with only a few mouse clicks or an API call, often with no downtime. Many Amazon RDS engine types allow you to launch one or more Read Replicas to offload read traffic from your primary database instance.

❍**Available and durable:** Amazon RDS runs on the same highly reliable infrastructure used by other Amazon Web Services. When you provision a Multi-AZ DB Instance, Amazon RDS synchronously replicates the data to a standby instance in a different Availability Zone (AZ). Amazon RDS has many other features that enhance reliability for critical production databases, including automated backups, database snapshots, and automatic host replacement.

❍**Fast:** Amazon RDS supports the most demanding database applications. You can choose between two SSD-backed storage options: one optimized for high-performance OLTP applications and the other for cost-effective general-purpose use. In addition, Amazon Aurora provides performance on par with commercial databases at 1/10th the cost.

❍**Secure:** Amazon RDS makes it easy to control network access to your database. Amazon RDS also lets you run your database instances in Amazon Virtual Private Cloud (Amazon VPC), enabling you to isolate your database instances and connect to your existing IT infrastructure through an industry-standard encrypted IPsec VPN. Many Amazon RDS engine types offer encryption at rest and encryption in transit.

❍**Inexpensive:** You pay very low rates and only for the resources you consume. In addition, you benefit from the option of On-Demand pricing with no up-front or long-term commitments or even lower hourly rates via our Reserved Instance pricing.

❍**Ease of use.** Admins don't need to learn specific database management tools, and they also can manage multiple database instances using the management console. RDS is compatible with database engines that users may already be familiar with, such as MySQL and Oracle, and it automates manual backup and recovery processes.

❍**Cost-effectiveness.** According to AWS, customers only pay for what they use. Also, the time spent maintaining instances is reduced because maintenance tasks, such as backups and patching are automated.

https://www.javatpoint.com/amazon-relational-database-service

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Amazon RDS: The Best Relational Database Service?

Companies these days are handling more data than ever: an average of 163 terabytes (163,000 gigabytes), according to a survey by IDG. Efficiently storing, processing and analyzing this data is essential in order to glean valuable insights and make informed business decisions.

Yet the question remains: What is the best way to store enterprise data? For many use cases, the most appealing choice is a relational database.

In addition to Microsoft Azure and Google Cloud Platform, Amazon Web Services (AWS) is one of the most popular public cloud computing platforms. AWS includes support for a suite of database services, one of which is Amazon RDS, aka Amazon Relational Database Service.

This article will discuss everything you need to know about Amazon RDS: its architecture, functionality, top features, top alternatives, and how Integrate.io can help you use Amazon RDS to its full potential.

Architecture: Amazon RDS and Amazon AWS

The purpose of a database is to store large quantities of information in an organized fashion. You can classify databases depending on how they structure and process this information.

Relational databases, of course, function with a relational model. The relational model is a method of structuring information that uses tables with columns and rows, similar to what you see in a spreadsheet in Microsoft Excel.

Each row in a relational database represents a record in that database, while each column represents a new piece of information about that record. For example, imagine a relational database that contains information about the students enrolled at a university. The rows of the database represent the different students, while the columns represent different data points, such as the student’s ID number, major, and GPA.

Providers contrast relational databases with non-relational databases (also known as NoSQL databases), which use a different model. Some common types of non-relational databases are key-value stores, document stores, and graph databases.

Amazon AWS offers a wide variety of services, including its own relational database engine known as Amazon Aurora and, of course, Amazon RDS. But Amazon RDS is not a database. It is a cloud computing solution from Amazon Web Services that aims to facilitate setting up, deploying, and scaling a relational database in the cloud.

First introduced in October 2009, Amazon RDS includes support for many of the most popular relational database solutions, including Oracle, Microsoft SQL Server, and PostgreSQL. As a fully managed service, Amazon RDS handles many of the onerous tasks of database management, such as migration, patching, and backup and recovery.

Like any other managed service in the cloud, the cloud provider (in this case, AWS) is responsible for provisioning the infrastructure and performing the tasks of maintenance and management. The customer (in this case, you) is only responsible for creating, managing, configuring, and deleting Amazon RDS instances as you deem necessary.

Users utilize the AWS command-line interface and management console along with the RDS API (Application Programming Interface) to create a database 'instance' or DB instance. A DB instance is a particular database 'environment' set up by the user. It can include more than one database, and when users apply settings to one instance they can automatically apply them across all the included databases. A manager can configure these instances for a specific CPU and storage usage.

Users can also set up 'read replicas.' These function as DB instances, but offer exclusively read-only functionality. You can use read replicas to offload read-only traffic from the principal or 'master' DB instance. If you set up Multi-AZ (availability zone), you may also set up read replicas in a different AZ from the principal database, meaning a different physical location. This can improve performance for certain locations. Finally, you can promote a read replica to the primary instance, typically with less than 30 seconds of delay, if the master DB experiences failover. This avoids problematic downtime.

Top Amazon RDS Alternatives

While Amazon RDS can be a strong offering for users who want a managed relational database in the cloud, it is not the only alternative out there.

Besides Amazon RDS, solutions such as MySQL, PostgreSQL, and MariaDB are all highly popular ways for businesses to store and manage their enterprise data. However, with so many different options for building a relational database, these solutions do not always come with a built-in way to integrate data from multiple sources, which is essential in order to gain deeper insights for your business needs.

The good news is you can easily integrate Amazon RDS with other data sources by using a third-party solution such as Integrate.io. With just a few clicks, Integrate.io enables data integration in the cloud between Amazon RDS and other relational databases. After that, you can process and analyze the combined data and then store the results back in your Amazon RDS database.

MySQL

MySQL is another managed database service that enables users to deploy cloud-native applications. It is open-source and originally developed by Oracle. MySQL is one of the most popular solutions on the market. The software tends to be extremely fast and is a robust solution, although it does not deliver the same level of ease of use as Amazon RDS. It is highly scalable and you can use it with various operating systems.

PostgreSQL

PostgreSQL is also open-source. It is an object-relational database system and is ACID-compliant (Atomicity, Consistency, Isolation and Durability). PostgreSQL is compatible with a variety of data types and formats and has various disaster recovery methods. The system supports synchronous, asynchronous and logical replication. There are different administration tools available, both free and paid. The main benefits of PostgreSQL are that it is risk-tolerant and low-maintenance and the source code is freely available. Many functions can, however, be very complex, and its speed is lacking compared to MySQL.

MariaDB

From the creators of MySQL, MariaDB is a fork from MySQL, developed as an alternative after Oracle bought MySQL. It performs replication faster. MariaDB is community-developed and supported, though the MariaDB Foundation primarily leads in documentation and developing the system. The entire purpose is to be as transparent and open as possible in development.

Integrate.io helps to monitor and analyze your data. The software can integrate with any of the solutions mentioned and brings you an additional level of ease with out-of-box data transformations.

The primary reason many businesses choose Amazon RDS over these other options is because of the AWS suite of tools that comes with it and the AWS management console. Other reasons are the system's enhanced availability, excellent durability, easy scalability, pricing model, and ease of managing individual database environments. Add to that robust documentation and tutorials managed and offered by Amazon itself.

Amazon RDS and Integrate.io

AWS RDS is a feature-rich and mature offering that makes it easy for even less tech-savvy users to operate their own relational databases. Integrate.io can help you gain the most benefit from your Amazon RDS deployment.

Integrate.io offers comprehensive data analysis and management and can integrate your Amazon RDS database with dozens of other services and products. In this way, you can use the technology stack that gives your business the greatest advantage. To learn more about how Integrate.io can benefit your business, get in touch to begin your journey toward simpler and more-efficient data integration.

https://www.integrate.io/blog/amazon-rds-what-is-it-and-how-does-it-work/#architecture

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